

Hands-On: Conditional Rendering using Switch Statement in React

In this lab, you will create a simple React component that uses a switch statement to conditionally render content based on a string value in the component's state.

Step 1: Create a new React component

Create a new file called `TrafficLight.js` and add the following code:

```
import React, { Component } from 'react';

class TrafficLight extends Component {
  state = {
    color: 'red',
  };

  render() {
    const { color } = this.state;

    let bulb;

    switch (color) {
      case 'red':
        bulb = <div className="red"></div>;
        break;
      case 'yellow':
        bulb = <div className="yellow"></div>;
        break;
      case 'green':
        bulb = <div className="green"></div>;
        break;
      default:
        bulb = null;
    }

    return (
      <div>
        {bulb}
        <button onClick={() => this.setState({ color: 'red' })}>Red</button>
        <button onClick={() => this.setState({ color: 'yellow' })}>Yellow</button>
        <button onClick={() => this.setState({ color: 'green' })}>Green</button>
      </div>
    );
  }
}
```

`export default TrafficLight;`

This component has an initial state with a string value to track the color of the traffic light. The `render` method uses a switch statement to conditionally render a div with the appropriate class

based on the `color` state. It also renders three buttons that update the state when clicked to simulate changing the traffic light color.

Step 2: Render the component

Create a new file called `App.js` and add the following code:

```
import React from 'react';
import TrafficLight from './TrafficLight';

function App() {
  return (
    <div className="App">
      <h1>Conditional Rendering using Switch Statement in React</h1>
      <TrafficLight />
    </div>
  );
}

export default App;
```

This component simply renders the `TrafficLight` component.

Step 3: Run the app

Run the app using `npm start` and test the `TrafficLight` component by clicking the buttons to change the color of the traffic light.

Congratulations! You have successfully implemented conditional rendering using a switch statement in a React component.